

Sphere $1 \rightarrow n$ Amplitudes in the $c = 1$ String

A numerical worldsheet note

24 August 2026

Abstract

This note records the normalization, soft-energy factorization, contour prescription, and numerical evidence for the sphere $1 \rightarrow 3$, $1 \rightarrow 4$, and $1 \rightarrow 5$ tachyon amplitudes in the $c = 1$ string. All three calculations are performed directly from the worldsheet on the imaginary-energy ray $\omega = it$. For $1 \rightarrow 3$, all thirty production points lie in the subtraction-free and residue-free chamber $0 < t < 1/2$: the original sixteen-point scan is augmented by fourteen interlaced points computed with the same numerical design. For $1 \rightarrow 4$, the original eighteen-point table is augmented by twelve new worldsheet evaluations to give thirty stored points. Twenty-nine points through $t = 0.48$ enter the primary quadratic fit; the inherited $t = 0.49$ near-second-wall point remains a diagnostic. The crossed DOZZ residue is included above the first Liouville-momentum wall, and the merged scan and worldsheet-only fits are frozen before the matrix-model comparison. For $1 \rightarrow 5$, the calculation is restricted to the residue-free chamber $0 < t < 1/3$; its original eight-point scan and subsequent eight-point local extension are merged and frozen before the matrix-model formula is imported. The audited data support

$$\mu^2 \mathcal{A}_{1 \rightarrow 3}^{\text{tree}}(\omega) = 3i\omega^4(1 + 3i\omega),$$

$$\mu^3 \mathcal{A}_{1 \rightarrow 4}^{\text{tree}}(\omega) = 8i\omega^5(1 + 2i\omega)(1 + 4i\omega)$$

and

$$\mu^4 \mathcal{A}_{1 \rightarrow 5}^{\text{tree}}(\omega) = 5i\omega^6(1 + 5i\omega)(2 + 5i\omega)(3 + 5i\omega).$$

We also state the expected $1 \rightarrow n$ formula and outline a scalable momentum-integration strategy. The arbitrary-multiplicity formula remains a comparison target rather than a general worldsheet proof.

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1 Scope and status

There are three numerical claims and one general extrapolation in this note.

1. The sphere $1 \rightarrow 3$ amplitude has been recomputed at thirty points with $0.16 \leq t \leq 0.46$. Fourteen new interlaced values were merged with the original frozen sixteen-point table without discarding any known datum. The raw convergent worldsheet correlator and its affine fits were frozen before the separate matrix-model comparison. Independent deep-scramble runs and changes of block and momentum quadrature parameters audit the quoted result.
2. The sphere $1 \rightarrow 4$ amplitude now has a thirty-point stored table: twelve new values are merged with all eighteen known values. The primary target-free quadratic fit uses the twenty-nine points through $t = 0.48$; the known $t = 0.49$ near-wall diagnostic is retained but excluded. The calculation has been checked against changes of momentum quadrature, Virasoro-block order, quasi-Monte Carlo depth, and OPE channel. Above the first contour wall, the required DOZZ residue is included. The merged worldsheet table and fits were hash-frozen before the separate matrix-model comparison.
3. The sphere $1 \rightarrow 5$ amplitude has been evaluated at sixteen points in the residue-free chamber $0 < t < 1/3$: an original frozen eight-point core and an independent eight-point local extension. The merged worldsheet scan was frozen again before its sixteen-point comparison. A mixed comb-star atlas keeps all three plumbing parameters in controlled channels, and a three-dimensional Liouville-momentum quadrature is used at every moduli point.
4. The compact formula for arbitrary $1 \rightarrow n$ scattering remains the analytic expectation suggested by target-space considerations. The $n = 3$, $n = 4$, and $n = 5$ cases now have direct numerical worldsheet evidence, but a proof for unrestricted n has not been supplied.

This distinction matters: numerical samples can identify a low-degree rational or polynomial answer only after one supplies an independent bound on its analytic form.

2 Common worldsheet normalization

Let I_{n+1} denote the labelled sphere matter integral for one incoming and n outgoing tachyons. In the BRY normalization,

$$\mathcal{A}_{1 \rightarrow n}^{\text{tree}} = i(g_s^{\text{BRY}})^{n+1} C_{S^2} I_{n+1}, \quad C_{S^2} = \frac{2\pi}{(g_s^{\text{BRY}})^2}. \quad (1)$$

Using

$$2\pi g_s^{\text{BRY}} = \mu^{-1}, \quad (2)$$

one obtains

$$\boxed{\mu^{n-1} \mathcal{A}_{1 \rightarrow n}^{\text{tree}} = i(2\pi)^{2-n} I_{n+1}.} \quad (3)$$

There is no factor of $1/n!$ in (3): the outgoing insertions are labelled before the equal-energy specialization is imposed.

After extracting the universal external soft factors, define a stripped quantity Q_n by

$$\boxed{\mu^{n-1} \mathcal{A}_{1 \rightarrow n}^{\text{tree}} = i\omega_{\text{in}} \prod_{a=1}^n \omega_a Q_n(\omega_{\text{in}}; \omega_1, \dots, \omega_n).} \quad (4)$$

No matrix-model expression is needed to define either I_{n+1} or Q_n . In each process-specific section, the target-space formula is therefore postponed until after the worldsheet integrand, imaginary-energy integration, and frozen numerical results have been presented.

3 Why the external powers of energy can be divided out

For the delta-normalized $c = 25$ Liouville structure constant,

$$C(P_1, P_2, P_3) \propto \prod_{j=1}^3 2P_j \Upsilon_1(1 + 2iP_j), \quad \Upsilon_1(1) = 1. \quad (5)$$

An external soft limit therefore contributes

$$C \sim 2P_{\text{ext}} = \omega_{\text{ext}}. \quad (6)$$

In a trivalent sphere decomposition, every external leg occurs in exactly one three-point coefficient. Provided that the remaining moduli and Liouville-momentum integrals contain no compensating infrared pole, the full amplitude contains the product

$$\omega_{\text{in}} \prod_{a=1}^n \omega_a. \quad (7)$$

For equal-energy $1 \rightarrow 4$ scattering, this factor is

$$(4\omega)\omega^4 = 4\omega^5. \quad (8)$$

Thus the factor ω^4 comes from the four outgoing external legs; the fifth soft factor, 4ω , comes from the incoming leg. This is a *multiplicative factorization*. It is not the subtraction of an ω^4 term from the amplitude. The caveat about a possible compensating infrared pole is precisely why the Liouville momentum integral must be controlled before using this division.

4 The sphere $1 \rightarrow 3$ amplitude

4.1 Worldsheet moduli integrand and normalization

For equal outgoing energies,

$$(\omega_{\text{in}}; \omega_1, \omega_2, \omega_3) = (3\omega; \omega, \omega, \omega), \quad (9)$$

the labelled normalization and stripped function are

$$\mu^2 \mathcal{A}_{1 \rightarrow 3}^{\text{tree}} = \frac{iI_4}{2\pi}, \quad Q_3(\omega) = \frac{I_4}{6\pi\omega^4}, \quad \mu^2 \mathcal{A}_{1 \rightarrow 3}^{\text{tree}} = 3i\omega^4 Q_3(\omega). \quad (10)$$

The worldsheet integral has one complex modulus and one internal Liouville momentum. In an s -channel frame it is

$$I_4(\omega) = \int_{\mathbb{C}} d^2z \int_0^\infty \frac{dP}{\pi} C\left(\frac{3\omega}{2}, \frac{\omega}{2}, P\right) C\left(\frac{\omega}{2}, \frac{\omega}{2}, P\right) \times \mathcal{I}_{X^0}^{(4)}(z, \bar{z}; 3\omega, -\omega, -\omega, -\omega) \left| \mathcal{F}_{25}^{(4)}(P; z) \right|^2. \quad (11)$$

Here $\mathcal{I}_{X^0}^{(4)}$ includes the timelike free-boson and gauge-fixed ghost powers, and $\mathcal{F}_{25}^{(4)}$ is the $c = 25$ Virasoro block in the chosen crossing frame. The momentum coefficient is

$$C\left(\frac{3\omega}{2}, \frac{\omega}{2}, P\right) C\left(\frac{\omega}{2}, \frac{\omega}{2}, P\right), \quad P \in [0, \infty), \quad (12)$$

with measure dP/π fixed by the delta-normalized Liouville states.

4.1.1 Convergent chamber and exact three-chart proposal

On $\omega = it$, the first pair of DOZZ poles capable of pinching the quotient contour is

$$P_{\pm} = \pm(2\omega - i), \quad (13)$$

so the first residue wall is $t = 1/2$. Near any of the three boundary divisors $z = 0, 1, \infty$, the leading local radial behavior at fixed P is

$$d^2z |z|^{-2+2t^2+2P^2} \sim dr r^{-1+2t^2+2P^2} d\theta. \quad (14)$$

It is integrable for every production value $t > 0$. Therefore the $1 \rightarrow 3$ imaginary-ray computation uses the raw correlator: no BRY finite part, collar subtraction, or residue term is present.

The moduli proposal is an equal mixture of power-law discs around the three cusps. With radial power $a = 2t^2$, its exact density with respect to d^2z is

$$p_a(z) = \frac{a}{6\pi} \left[\mathbf{1}_{|z|<1} |z|^{a-2} + \mathbf{1}_{|1-z|<1} |1-z|^{a-2} + \mathbf{1}_{|z|>1} |z|^{-a-2} \right]. \quad (15)$$

Every sample is divided by the full mixture density, rather than only by the density of the chart that generated it. This makes the estimator unbiased on overlaps. Crossing covariance under $z \mapsto 1-z$ and $z \mapsto 1/z$ was checked pointwise at relative precision between 10^{-16} and 10^{-15} .

The production settings were block order 10, two Gauss-Legendre momentum panels with 24 nodes per panel, $P_{\max} = 8$, eight independent scrambled Sobol replicates, and 2^{13} moduli points per replicate. The imaginary part of the raw result is at most of relative machine precision throughout the scan.

4.2 Imaginary-energy worldsheet integration

The original sixteen-point worldsheet table was augmented by fourteen interlaced points

$$t = 0.17, 0.19, \dots, 0.43. \quad (16)$$

These fill the first fourteen gaps of the original $t = 0.16, 0.18, \dots, 0.46$ grid; the unused midpoint $t = 0.45$ is the one nearest the residue wall. Every known datum is retained. The extension uses the same block order, momentum quadrature, cutoff, Sobol depth, and replicate count as the original scan. Its seed stream begins after the sixteen known points, so its randomized scrambles are disjoint. The extension and merged table, respectively, have SHA-256 hashes

eaa6a8f131922efed3cdd6029fa83ec98fc9c50ed43c5b7a68a139428e5fa87f,
 ab785a09a9d57c68d1e84f12c7c738e83cfaae5fa9b4d7021afd2e4ac1e5a3b9.

The worldsheet-only affine fitter was then run and frozen, still without any matrix-model coefficient. The fit artifact has SHA-256 hash

c0acd1b43a9858c1187967b0ced75f6a383761b68a60fd9ff260e22be5fb274a.

The primary unweighted fit to all thirty production points is

$$Q_3(it) = 0.999774(87) - 2.999429(274)t, \quad \text{RMS} = 1.2655 \times 10^{-4}. \quad (17)$$

The fourteen new points alone give the sharper cohort diagnostic

$$Q_3(it) = 0.999958(76) - 2.999979(243)t, \quad \text{RMS} = 6.7992 \times 10^{-5}. \quad (18)$$

Replacing the five previously audited known points by independent 12×2^{14} deep-scramble evaluations gives the thirty-point sensitivity fit

$$Q_3(it) = 0.999900(50) - 2.999776(158)t, \quad \text{RMS} = 7.2742 \times 10^{-5}. \quad (19)$$

The uncertainties in parentheses come from the residual scatter of the corresponding worldsheet cohort, not from the matrix-model curve.

4.3 Post-freeze comparison with the matrix model

Only after the merged scan, worldsheet fits, and convergence audits were frozen was the comparison function evaluated. It is

$$Q_3^{\text{expected}}(\omega) = 1 + 3i\omega, \quad \boxed{\mu^2 \mathcal{A}_{1 \rightarrow 3}^{\text{tree}}(it) = 3it^4(1 - 3t)}. \quad (20)$$

The fourteen new points alone give $\chi_{\text{QMC}}^2 = 15.85/14$ against this curve, with maximum absolute pull 2.38 and RMS difference 7.66×10^{-5} . For the complete table, using the five independent deep values and assigning each a conservative error equal to the largest of the production error, deep-scramble error, and independent spread gives $\chi^2 = 35.56/30$, again with maximum pull 2.38. The raw production replicate errors alone give $\chi^2 = 66.18/30$ and maximum pull 5.16. This excess is inherited from the known $t = 0.32$ point, whose unusually small production replicate error was already shown to undercover the independent scramble fluctuation. The deep result there is $Q_3(0.32i) = 0.03999302(520)$, while changes of block order, momentum order, and cutoff move the answer by less than 2×10^{-9} .

Table 1: Frozen thirty-point sphere $1 \rightarrow 3$ worldsheet data. K and N denote the sixteen known and fourteen new points. The expected column was added only after the merged table and worldsheet-only fits were frozen.

t	set	Q_3^{WS}	$\sigma_{Q, \text{QMC}}$	Q_3^{expected}	$\text{Im}(\mu^2 \mathcal{A}^{\text{WS}})$
0.16	K	+0.519331219	2.93×10^{-4}	+0.520000	$+1.021046722 \times 10^{-3}$
0.17	N	+0.489879571	2.22×10^{-4}	+0.490000	$+1.227456951 \times 10^{-3}$
0.18	K	+0.459910868	1.88×10^{-4}	+0.460000	$+1.448388097 \times 10^{-3}$

continued on next page

Table 1 continued

t	set	Q_3^{WS}	$\sigma_{Q,\text{QMC}}$	Q_3^{expected}	$\text{Im}(\mu^2 \mathcal{A}^{\text{WS}})$
0.19	N	+0.430099451	1.00×10^{-4}	+0.430000	$+1.681529718 \times 10^{-3}$
0.20	K	+0.400084688	1.82×10^{-4}	+0.400000	$+1.920406501 \times 10^{-3}$
0.21	N	+0.369961806	1.93×10^{-4}	+0.370000	$+2.158516259 \times 10^{-3}$
0.22	K	+0.339818677	1.25×10^{-4}	+0.340000	$+2.388136920 \times 10^{-3}$
0.23	N	+0.309863956	1.02×10^{-4}	+0.310000	$+2.601379176 \times 10^{-3}$
0.24	K	+0.279826079	1.01×10^{-4}	+0.280000	$+2.785187318 \times 10^{-3}$
0.25	N	+0.249955713	6.79×10^{-5}	+0.250000	$+2.929168511 \times 10^{-3}$
0.26	K	+0.220072276	7.51×10^{-5}	+0.220000	$+3.017032448 \times 10^{-3}$
0.27	N	+0.189899868	8.45×10^{-5}	+0.190000	$+3.027617265 \times 10^{-3}$
0.28	K	+0.159940247	3.87×10^{-5}	+0.160000	$+2.949246968 \times 10^{-3}$
0.29	N	+0.130023447	4.02×10^{-5}	+0.130000	$+2.758893410 \times 10^{-3}$
0.30	K	+0.099992502	3.26×10^{-5}	+0.100000	$+2.429817789 \times 10^{-3}$
0.31	N	+0.069981942	1.26×10^{-5}	+0.070000	$+1.938893786 \times 10^{-3}$
0.32	K	+0.039966050	6.58×10^{-6}	+0.040000	$+1.257223232 \times 10^{-3}$
0.33	N	+0.009998147	3.69×10^{-6}	+0.010000	$+3.557103766 \times 10^{-4}$
0.34	K	-0.020000341	2.54×10^{-6}	-0.020000	$-8.018152530 \times 10^{-4}$
0.35	N	-0.049994040	1.54×10^{-5}	-0.050000	$-2.250669169 \times 10^{-3}$
0.36	K	-0.079970675	2.53×10^{-5}	-0.080000	$-4.029600775 \times 10^{-3}$
0.37	N	-0.109978317	3.15×10^{-5}	-0.110000	$-6.183512183 \times 10^{-3}$
0.38	K	-0.140003033	3.76×10^{-5}	-0.140000	$-8.757760933 \times 10^{-3}$
0.39	N	-0.170048919	3.58×10^{-5}	-0.170000	$-1.180194427 \times 10^{-2}$
0.40	K	-0.200066847	4.13×10^{-5}	-0.200000	$-1.536513382 \times 10^{-2}$
0.41	N	-0.229987890	3.60×10^{-5}	-0.230000	$-1.949672432 \times 10^{-2}$
0.42	K	-0.260080552	5.03×10^{-5}	-0.260000	$-2.427874840 \times 10^{-2}$
0.43	N	-0.290147842	6.20×10^{-5}	-0.290000	$-2.975873200 \times 10^{-2}$
0.44	K	-0.319929356	4.75×10^{-5}	-0.320000	$-3.597377819 \times 10^{-2}$
0.46	K	-0.379961069	4.34×10^{-5}	-0.380000	$-5.103776908 \times 10^{-2}$

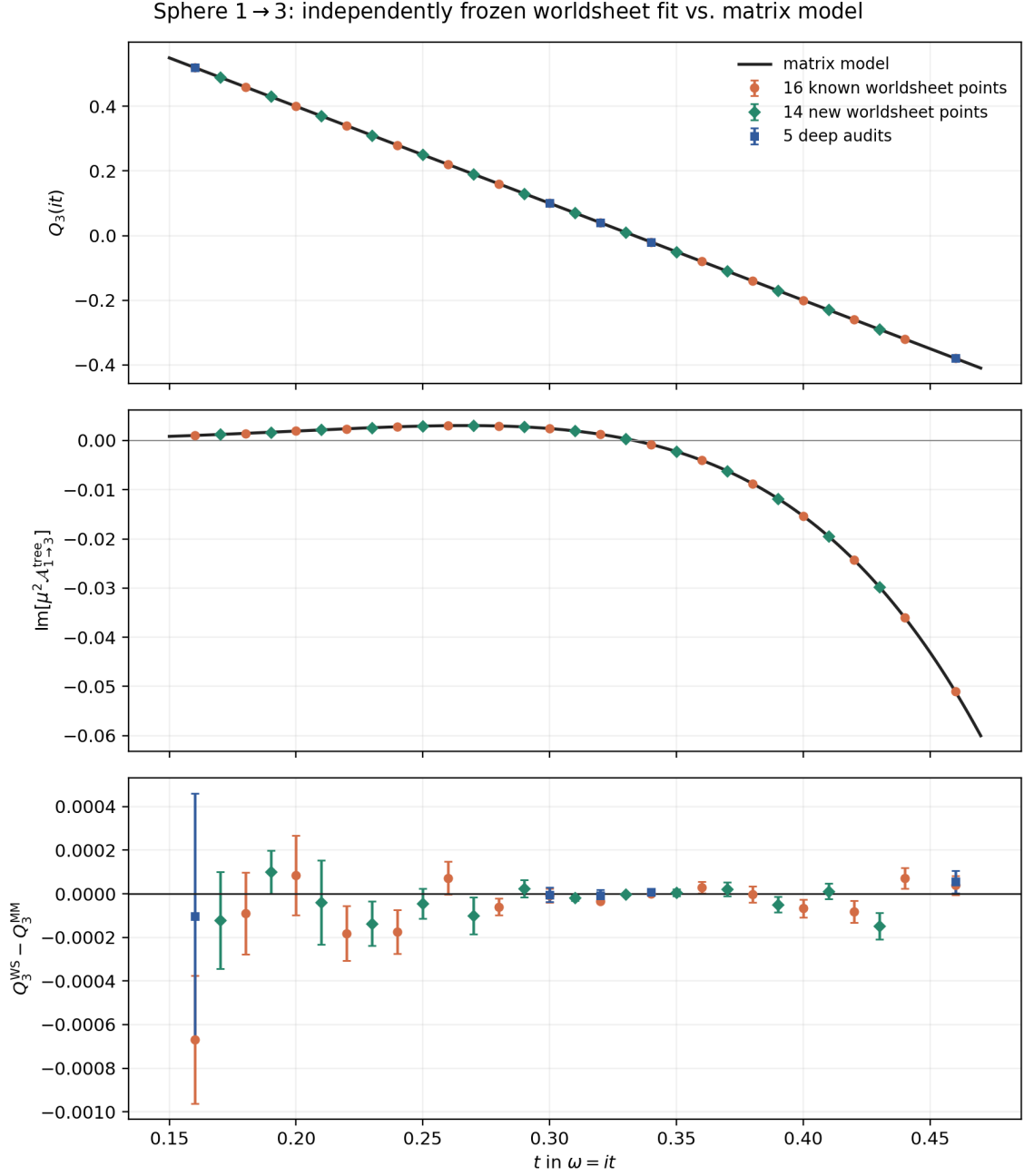


Figure 1: Sphere 1 $\rightarrow$ 3 on $\omega = it$. Orange circles are the sixteen known production points, green diamonds the fourteen new interlaced points, and blue squares the five independent deep audits with conservative errors. The black curve is the matrix-model expression evaluated only after the thirty-point scan and worldsheet fits were frozen. The panels show Q_3 , the imaginary part of the normalized amplitude, and the pointwise worldsheet-minus-matrix difference.

5 The sphere $1 \rightarrow 4$ amplitude

5.1 Worldsheet moduli integrand and normalization

For the equal-energy kinematics

$$(\omega_{\text{in}}; \omega_1, \omega_2, \omega_3, \omega_4) = (4\omega; \omega, \omega, \omega, \omega), \quad (21)$$

the two-complex-dimensional worldsheet integral in any chosen trivalent channel has the form

$$I_5(\omega) = \int_{\mathcal{M}_{0,5}} d\mu_5 \int_0^\infty \frac{dP_1}{\pi} \frac{dP_2}{\pi} \mathcal{I}_{X^0}^{(5)} \prod_{v=1}^3 C_L^{(v)} \left| \mathcal{F}_{25}^{(5)}(P_1, P_2; q_1, q_2) \right|^2. \quad (22)$$

Here $d\mu_5$ is the gauge-fixed moduli measure including the chart Jacobian, the three $C_L^{(v)}$ are the delta-normalized Liouville structure constants, and $\mathcal{I}_{X^0}^{(5)}$ is the timelike factor for signed energies $(4\omega, -\omega, -\omega, -\omega, -\omega)$. We define

$$Q_4(\omega) = \frac{I_5}{16\pi^2\omega^5}. \quad (23)$$

The normalized amplitude is then

$$\mu^3 \mathcal{A}_{1 \rightarrow 4}^{\text{tree}} = \frac{iI_5}{4\pi^2} = 4i\omega^5 Q_4(\omega). \quad (24)$$

5.2 Imaginary-energy worldsheet integration

The direct integration is performed on $\omega = it$. The conversion from the stripped worldsheet result to the normalized amplitude follows from (24) alone:

$$\mu^3 \mathcal{A}(it) = -4t^5 Q_4(it), \quad \sigma_A = 4t^5 \sigma_Q. \quad (25)$$

5.2.1 Numerical values

The frozen eighteen-point table was augmented by twelve direct worldsheet evaluations at

$$t = 0.19, 0.21, 0.23, 0.27, 0.29, 0.31, 0.33, 0.35, 0.37, 0.39, 0.44, 0.47. \quad (26)$$

Every known point is retained, including the special probes at $t = 0.249$ and 0.251 and the $t = 0.49$ diagnostic. The new design avoids the exact structure-constant zero at $t = 1/4$, the first contour wall at $t = 2/5$, and the second wall at $t = 1/2$. All new points use momentum order 20, $P_{\text{max}} = 6$, the threshold-adapted rule $P = 6u^{5/4}$, four independent scrambled Sobol replicates, and the c -recursion block. The extension and merged worldsheet table have SHA-256 hashes

```
be9af08cbd871f727ef6068167fb27467f18428013eaa6a9b9d2fee8e03c584c,
63e8df62c1301110f4131b846dc2908a18a62500564f5a0a891dd90d01f360d1.
```

The full thirty-point data are given in Table 2. K and N denote the eighteen known and twelve new points. The expected column was evaluated only after the merged table and target-free fits were frozen. Quoted standard errors are QMC replicate errors; the independent block-order spreads at three representative new points enter only the later conservative comparison.

Table 2: Frozen thirty-point sphere $1 \rightarrow 4$ dataset. The final row is the near-second-wall diagnostic and is excluded from the primary fit.

t	set	$Q_4^{\text{WS}}(it)$	$\sigma_{Q,\text{QMC}}$	Q_4^{expected}	$\mu^3 \mathcal{A}^{\text{WS}}(it)$	Prescription
0.18	K	+0.360141905	3.59×10^{-4}	+0.358400	$-2.722050477 \times 10^{-4}$	Real
0.19	N	+0.297868413	7.69×10^{-4}	+0.297600	$-2.950206718 \times 10^{-4}$	Real
0.20	K	+0.240682960	2.83×10^{-4}	+0.240000	$-3.080741891 \times 10^{-4}$	Real
0.21	N	+0.185361345	2.67×10^{-4}	+0.185600	$-3.028137817 \times 10^{-4}$	Real
0.22	K	+0.134607326	1.68×10^{-4}	+0.134400	$-2.774866497 \times 10^{-4}$	Real
0.23	N	+0.086355703	1.24×10^{-4}	+0.086400	$-2.223259708 \times 10^{-4}$	Real
0.24	K	+0.041592765	4.09×10^{-5}	+0.041600	$-1.324750207 \times 10^{-4}$	Real
0.249	K	+0.004014571	3.94×10^{-6}	+0.004016	$-1.537077710 \times 10^{-5}$	Real
0.251	K	-0.003982449	3.91×10^{-6}	-0.003984	$+1.587006931 \times 10^{-5}$	Real
0.26	K	-0.038411505	5.16×10^{-5}	-0.038400	$+1.825526146 \times 10^{-4}$	Real
0.27	N	-0.073604833	1.94×10^{-4}	-0.073600	$+4.224595627 \times 10^{-4}$	Real
0.28	K	-0.105602511	1.49×10^{-4}	-0.105600	$+7.269832315 \times 10^{-4}$	Real
0.29	N	-0.134252025	2.00×10^{-4}	-0.134400	$+1.101465313 \times 10^{-3}$	Real
0.30	K	-0.159973349	2.38×10^{-4}	-0.160000	$+1.554940955 \times 10^{-3}$	Real
0.31	N	-0.182339671	3.06×10^{-4}	-0.182400	$+2.088091993 \times 10^{-3}$	Real
0.32	K	-0.201535953	3.18×10^{-4}	-0.201600	$+2.704969768 \times 10^{-3}$	Real
0.33	N	-0.216868021	5.17×10^{-4}	-0.217600	$+3.394886091 \times 10^{-3}$	Real
0.34	K	-0.230229797	2.51×10^{-4}	-0.230400	$+4.184235372 \times 10^{-3}$	Real
0.35	N	-0.242549497	1.40×10^{-3}	-0.240000	$+5.095661747 \times 10^{-3}$	Continued; inactive
0.36	K	-0.247439049	1.71×10^{-3}	-0.246400	$+5.984677245 \times 10^{-3}$	Continued; inactive
0.37	N	-0.250033136	1.44×10^{-3}	-0.249600	$+6.935314814 \times 10^{-3}$	Continued; inactive
0.38	K	-0.250784365	1.82×10^{-3}	-0.249600	$+7.948376520 \times 10^{-3}$	Continued; inactive
0.39	N	-0.247485294	1.18×10^{-3}	-0.246400	$+8.931664969 \times 10^{-3}$	Continued; inactive
0.395	K	-0.244256937	1.85×10^{-3}	-0.243600	$+9.394904648 \times 10^{-3}$	Continued; inactive
0.42	K	-0.218868687	1.74×10^{-3}	-0.217600	$+1.144168732 \times 10^{-2}$	Residue corrected
0.44	N	-0.182612916	6.71×10^{-4}	-0.182400	$+1.204633301 \times 10^{-2}$	Residue corrected
0.46	K	-0.135603621	1.17×10^{-3}	-0.134400	$+1.117173014 \times 10^{-2}$	Residue corrected
0.47	N	-0.105739056	8.75×10^{-4}	-0.105600	$+9.700289781 \times 10^{-3}$	Residue corrected
0.48	K	-0.073467644	6.60×10^{-4}	-0.073600	$+7.487938910 \times 10^{-3}$	Residue corrected
0.49	K	-0.036352662	4.19×10^{-4}	-0.038400	$+4.107490899 \times 10^{-3}$	Residue; diagnostic

5.2.2 Liouville contour continuation

On $\omega = it$, the incoming–outgoing cherry coefficient has the nearest pair of poles

$$P_{\pm} = \pm \left(\frac{5}{2}\omega - i \right). \quad (27)$$

They pinch the endpoint of the real quotient contour at

$$t = \frac{2}{5}. \quad (28)$$

For $2/5 < t < 1/2$, the continued contribution from an internal edge whose cherry contains both the incoming operator and one outgoing operator is

$$\int_0^\infty \frac{dP}{\pi} I(P) - 2i \operatorname{Res}_{P=(5/2)\omega-i} I(P). \quad (29)$$

Channels in which the incoming operator sits at the middle vertex receive no residue at this first wall.

The points at $t = 0.35, 0.36, 0.37, 0.38, 0.39, 0.395$ were evaluated using the same continued-contour implementation, but the residue term is inactive below $t = 2/5$. Every stored point above the wall includes the residue term in (29).

A second pole family pinches at $t = 1/2$. The first-wall formula cannot be continued past that value without adding the next residue stratum. The $t = 0.49$ point is therefore retained as a diagnostic rather than treated as a production point.

5.2.3 Worksheet numerical audit

The following checks concern worksheet convergence only. Their computation routines do not evaluate or import a matrix-model amplitude.

1. At $t = 0.34$, momentum orders 20 and 24 gave -0.2303047302 and -0.2303034636 , respectively.
2. At the same point, varying $P_{\max} = 5, 6, 7$ produced a range of only 2.6×10^{-5} .
3. Sobol powers 9, 10, and 11 agreed within their QMC errors.
4. At $t = 0.26$, block levels 8 and 10 gave $-0.03849398(20)$ and $-0.03836665(22)$; their spread is retained as a block-truncation diagnostic.
5. An equal-quality overlapping-channel test at $t = 0.34$ improved from a 0.44% mismatch at block level 6 to a 0.069% mismatch at level 8.

6. Three independent new-point block-order audits were performed after the primary fit was frozen. At $t = 0.31$, the production block-6 value -0.18233967 and block-8 audit -0.18240429 differ by 6.46×10^{-5} . At $t = 0.35$, the block-8 production value -0.24254950 and block-6 audit -0.23855076 differ by 4.00×10^{-3} . At the residue-corrected point $t = 0.44$, the corresponding values -0.18261292 and -0.18098326 differ by 1.63×10^{-3} . The latter two spreads exceed the replicate errors and are carried as conservative local uncertainties in the downstream comparison. The frozen audit has SHA-256 hash

3b3a966dcf215237baae09d5014f47dff735dc2e28c1bb67c68309e8b6a73ce5.

7. Independent descendant, h -recursion, c -recursion, atlas, Jacobian, and contour-wall checks passed.

The stored points continue to bracket the first exact zero without evaluating the logarithmic DOZZ expression at its pole:

$$Q_4(i0.249) = 0.00401457(39), \quad Q_4(i0.251) = -0.00398245(39). \quad (30)$$

At exactly $t = 1/4$, the evaluator encounters the corresponding structure-constant zero, so that coordinate is intentionally absent from the numerical design.

5.2.4 Target-free quadratic fit on the imaginary ray

For analytic continuation from the imaginary-energy chamber, we impose the explicit degree-two ansatz

$$Q_4(it) = a + bt + ct^2. \quad (31)$$

A mechanical extraction first writes a points-only table containing no matrix-model coefficient. Its SHA-256 hash is

611d5f1f75d7fdd0240aaa134803b656f6a157c3928a91cca0e6f4cc85c00a12.

The target-free fitting program uses all twenty-nine points through $t = 0.48$, including the five residue-corrected values at $t = 0.42, 0.44, 0.46, 0.47, 0.48$. The $t = 0.49$ near-second-wall diagnostic is retained in the table but excluded from the primary fit. Because the stored pointwise errors do not supply the full correlated block/discretization covariance, the primary fit is unweighted ordinary least squares. Its frozen artifact has SHA-256 hash

5e165ec33cdbc721af8ce5550212315eb41a07323bad42b7bd83ae7f51e61907.

The result is

$$(a, b, c) = (2.004567 \pm 0.001687, -12.026247 \pm 0.010763, 16.032869 \pm 0.016273), \quad (32)$$

where the parenthesized errors come from the residual scatter. The RMS residual is 6.04×10^{-4} and the largest absolute residual is 1.96×10^{-3} . The twelve new points alone give the independent cohort fit

$$(a, b, c)_{\text{new } 12} = (2.002855 \pm 0.003581, -12.017837 \pm 0.022820, 16.023398 \pm 0.034654), \quad (33)$$

with RMS residual 7.47×10^{-4} . The roots of the primary frozen fit are $t = 0.25001494$ and 0.50008459 ; the latter is quoted only as the root of the degree-two reconstruction because the first-wall contour formula is not validated at the second wall.

5.3 Post-freeze comparison with the matrix model

Only after the points-only table and the fit in (32) were written did the downstream program evaluate the matrix-model coefficients. The comparison target is

$$Q_4^{\text{expected}}(\omega) = 2(1 + 2i\omega)(1 + 4i\omega), \quad Q_4^{\text{expected}}(it) = 2(1 - 2t)(1 - 4t). \quad (34)$$

Consequently,

$$\boxed{\mu^3 \mathcal{A}_{1 \rightarrow 4}^{\text{tree}}(\omega) = 8i\omega^5(1 + 2i\omega)(1 + 4i\omega),} \quad \mu^3 \mathcal{A}_{1 \rightarrow 4}^{\text{tree}}(it) = -8t^5(1 - 2t)(1 - 4t). \quad (35)$$

For the twelve new points, the QMC-only pointwise statistic is $\chi^2 = 8.01/12$, with maximum absolute pull 1.82. Enlarging the errors at the three audited new points to the largest of the production QMC error, audit QMC error, and independent block-order spread gives $\chi^2 = 5.00/12$ and maximum pull 1.42. The complete twenty-nine-point primary table gives $\chi^2 = 42.35/29$ using QMC errors and 39.33/29 with the same conservative audit prescription. Its largest pull, 4.86, is the inherited known point at $t = 0.18$, not a new datum.

The primary fitted coefficients differ from $(2, -12, 16)$ by 0.228%, 0.219%, and 0.205%, respectively. Its reconstructed roots 0.25001494 and 0.50008459 agree with the matrix-model roots $1/4$ and $1/2$ at the stated numerical resolution. Agreement over a finite set of points does not by itself prove the quadratic degree; that degree is the explicit analytic ansatz in (31). This comparison is an imaginary-energy analytic-continuation result and does not claim that the separate direct physical- $i\epsilon$ finite-part smoke test is complete.

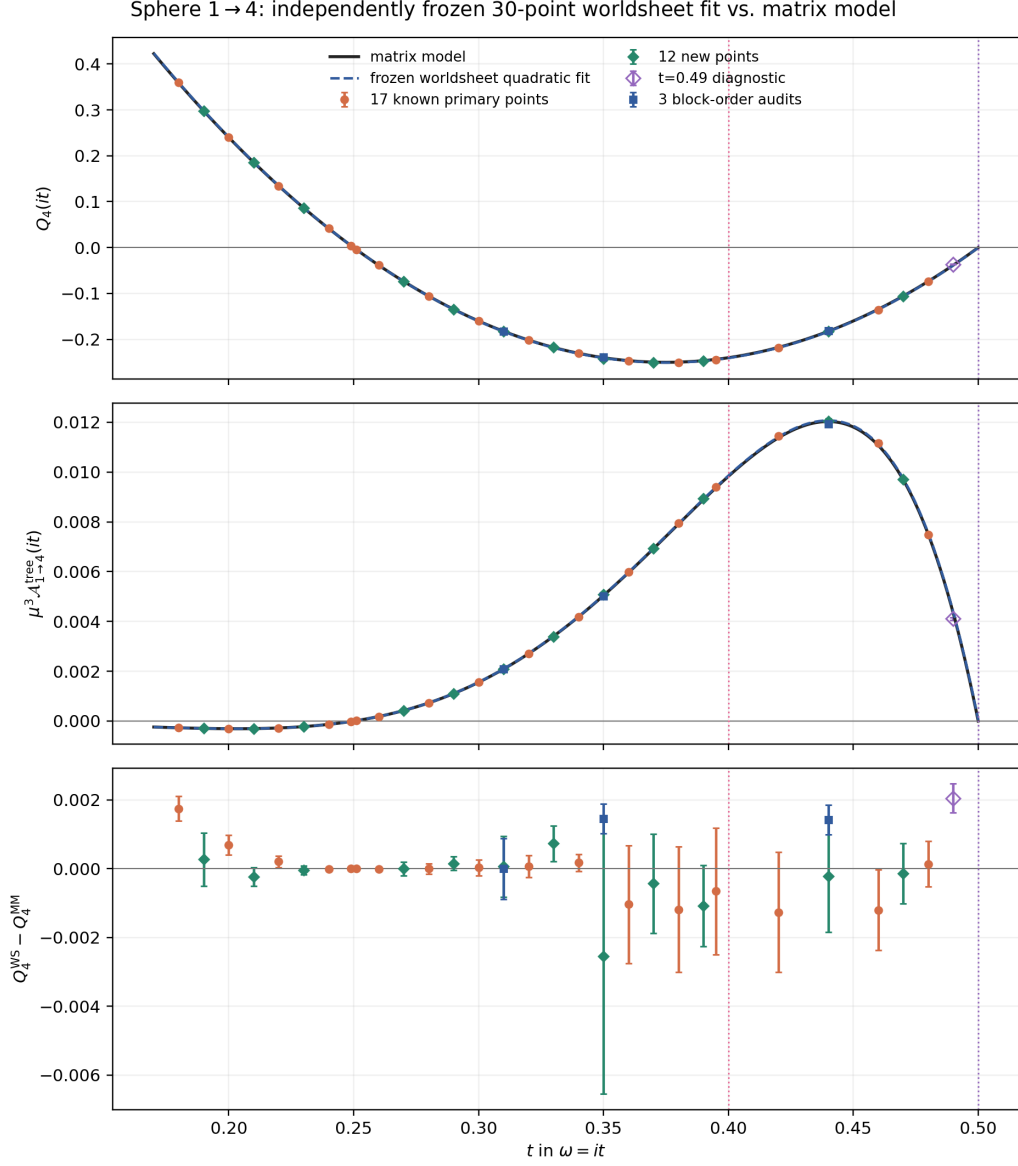


Figure 2: Sphere 1 $\rightarrow$ 4 on $\omega = it$. Orange circles are the seventeen known primary points, green diamonds the twelve new points, and the open purple diamond the inherited $t = 0.49$ near-wall diagnostic. Blue squares are the three independent block-order audits. The black curve is the matrix model evaluated after the thirty-point table and target-free fits were frozen; the dashed blue curve is the frozen twenty-nine-point worldsheet fit. The panels show Q_4 , the normalized amplitude, and the pointwise worldsheet-minus-matrix difference. Error bars on audited production points use the conservative spread described in the text.

6 The sphere $1 \rightarrow 5$ amplitude

6.1 Worldsheet moduli integrand and normalization

For the equal-energy kinematics

$$(\omega_{\text{in}}; \omega_1, \dots, \omega_5) = (5\omega; \omega, \omega, \omega, \omega, \omega), \quad (36)$$

the six-point worldsheet integral in any fixed trivalent channel has the schematic form

$$I_6(\omega) = \int_{\mathcal{M}_{0,6}} d\mu_6 \int_0^\infty \prod_{j=1}^3 \frac{dP_j}{\pi} \mathcal{I}_{X^0}^{(6)} \prod_{v=1}^4 C_L^{(v)} \left| \mathcal{F}_{25}^{(6)}(P_1, P_2, P_3; q_1, q_2, q_3) \right|^2. \quad (37)$$

Here $d\mu_6$ is the gauge-fixed moduli measure including the selected-chart Jacobian, the four $C_L^{(v)}$ are delta-normalized Liouville structure constants, and $\mathcal{I}_{X^0}^{(6)}$ is the timelike factor for signed energies $(5\omega, -\omega, -\omega, -\omega, -\omega, -\omega)$. the labelled normalization (3) gives

$$\mu^4 \mathcal{A}_{1 \rightarrow 5}^{\text{tree}} = \frac{iI_6}{8\pi^3}. \quad (38)$$

The product of external soft factors is $5\omega^6$. It is therefore useful to define

$$Q_5(\omega) = \frac{I_6}{40\pi^3\omega^6}, \quad \mu^4 \mathcal{A}_{1 \rightarrow 5}^{\text{tree}} = 5i\omega^6 Q_5(\omega). \quad (39)$$

6.1.1 Mixed atlas and channel-dependent six-point block

The six-point moduli integral is split with an equal mixture of 720 oriented comb charts and 720 oriented star charts. After forgetting orientations, these represent 90 comb trees and 15 star trees. At every sampled point the code reconstructs all admissible plumbing coordinates and evaluates the Virasoro block in the topology with the smallest maximum plumbing radius. The production scan selected comb and star channels in nearly equal proportions; the largest selected radius was 0.9027.

The comb and star six-point blocks were implemented independently. Their global limits and their c -recursions agree with direct descendant definitions through the tested orders. The maximum relative discrepancy in the comb c -recursion tests was 1.2×10^{-15} , while the corresponding star discrepancy was 1.9×10^{-15} . Round trips between moduli and plumbing coordinates and the complex Jacobians agree at approximately 10^{-16} . A triple-cherry degeneration, which is poorly represented by a comb, is mapped to star radii as small as (0.1, 0.03, 0.01).

6.1.2 Three-dimensional Liouville-momentum integration

The three internal momenta are mapped as

$$P_j = P_{\text{max}} u_j^{5/4}, \quad 0 < u_j < 1, \quad P_{\text{max}} = 5. \quad (40)$$

Gauss–Legendre rules of adjacent orders (8, 9, 10) are used on the three edges. The unequal orders avoid accidental simultaneous Kac-pole collisions in the c -recursion. The DOZZ/Yin factors are evaluated logarithmically, and the complete $P_1 P_2 P_3$ contraction tensor is cached for reuse in the moduli integral. The production block order is four. Six independently scrambled Sobol replicates, with 128 moduli points per replicate and radial CDF $r^{0.2}$, determine the QMC error.

6.2 Imaginary-energy worldsheet integration

On the convergent ray $\omega = it$, the normalization above gives

$$Q_5(it) = -\frac{I_6}{40\pi^3 t^6}, \quad \mu^4 \mathcal{A}_{1 \rightarrow 5}^{\text{tree}}(it) = -5it^6 Q_5(it). \quad (41)$$

The nearest moving pole pair is

$$P_{\pm} = \pm(3\omega - i). \quad (42)$$

It reaches the real momentum contour at

$$t = \frac{1}{3}. \quad (43)$$

All production points satisfy $0 < t < 1/3$. No pole has crossed any real P_j contour in this chamber, so no residue term is present. This is an analytic statement about the contour, distinct from numerical control of the ordinary quadrature remainder. The implementation rejects input outside this chamber rather than silently omitting residues.

6.2.1 Blind freeze and numerical audit

The original eight-point worldsheet scan was completed before any matrix-model formula was loaded. Its frozen SHA-256 hash is

216925ad78f35331a39aad26d28a83a61bef415509ddfd6df491c127dd0e62a.

Eight further local points at

$$t = 0.155, 0.180, 0.195, 0.205, 0.210, 0.240, 0.280, 0.320 \quad (44)$$

were then computed by the same worldsheet-only driver. The original and extension tables were merged without changing any value and frozen before a new sixteen-point comparison. The merged SHA-256 hash is

fa8b60ce2bb1c34cec61ba7629f0051d15762d522d0527b5404fc1c4f36be0a6.

At the representative point $t = 0.18$, the following shifts in Q_5 were found:

$$(8, 9, 10) \text{ versus } (7, 8, 9) : 3.43 \times 10^{-4}, \quad (45)$$

$$\text{block order 4 versus 6} : 7.00 \times 10^{-4}, \quad (46)$$

$$P_{\text{max}} = 5 \text{ versus } 4 : 6.46 \times 10^{-4}. \quad (47)$$

Their quadrature sum defines the conservative absolute discretization error

$$\delta_{\text{disc}} Q_5 = 1.012 \times 10^{-3}. \quad (48)$$

An alternate radial sampler shifted Q_5 by 0.00729, within its combined QMC uncertainty of approximately 0.010. It therefore revealed no resolved importance-sampling bias. No regulated-block or direct-block fallback was used at any production point.

6.2.2 Target-free cubic fit on the imaginary ray

Before loading any comparison formula, a mechanical extraction wrote a points-only table from the frozen sixteen-point scan. Its SHA-256 hash is

eeb8243db391dbae895b52823b25bf93ac8168b962e31e79ab062302cb054d6b.

The fitting program contains no matrix-model coefficient. It assumes only the explicit analytic ansatz

$$Q_5(it) = a + bt + ct^2 + dt^3. \quad (49)$$

The primary generalized least-squares covariance treats the scrambled-Sobol QMC errors as independent and the single discretization estimate in (48) as one fully correlated additive nuisance across all t . The frozen target-free result is

$$\begin{aligned} (a, b, c, d) &= (6.33158, -58.91205, 164.86491, -142.96917), \\ \sigma_{a,b,c,d} &= (0.37817, 4.88690, 20.78288, 28.96791), \end{aligned} \quad (50)$$

with $\chi^2 = 1.13$ for 12 residual degrees of freedom. The low value is consistent with the deliberately conservative error model and should not be read as evidence that every systematic is statistically independent.

As conditioning diagnostics, the two alternative coefficient vectors are

$$\begin{aligned} (a, b, c, d)_{\text{diag}} &= (6.36015, -59.23433, 166.04757, -144.38683), \\ (a, b, c, d)_{\text{unweighted}} &= (7.05086, -67.66480, 199.65477, -188.22656). \end{aligned} \quad (51) \quad (52)$$

The first combines the discretization estimate diagonally, while the second is an ordinary unweighted fit. The coefficient vector is therefore less sharply determined than the fitted curve: all points lie in the narrow interval $0.14 \leq t \leq 0.32$, and the lowest- t points carry the largest errors. The correlated GLS prescription above is fixed as the primary result; these two alternatives are reported only as sensitivity tests.

6.3 Post-freeze comparison with the matrix model

Only after both the points-only table and the target-free cubic fit were frozen was the comparison program allowed to load

$$\boxed{Q_5^{\text{expected}}(\omega) = (1 + 5i\omega)(2 + 5i\omega)(3 + 5i\omega).} \quad (53)$$

Equivalently,

$$Q_5^{\text{expected}}(it) = (1 - 5t)(2 - 5t)(3 - 5t). \quad (54)$$

The comparison is summarized in Table 3. The quoted error is

$$\sigma_{Q,\text{combined}} = \sqrt{\sigma_{Q,\text{QMC}}^2 + (\delta_{\text{disc}} Q_5)^2}. \quad (55)$$

The worksheet calculation resolves the expected sign change without using the target formula:

$$Q_5(i0.199) = 0.010171(1048), \quad Q_5(i0.201) = -0.010007(1045), \quad (56)$$

Table 3: The original eight-point core of the frozen sphere $1 \rightarrow 5$ worldsheet data below the first residue wall. The matrix-model column was evaluated only after the worldsheet file was frozen.

t	Q_5^{WS}	$\sigma_{Q,\text{combined}}$	Q_5^{expected}	$\text{Im}(\mu^4 \mathcal{A}^{\text{WS}})$
0.140	0.982781	0.102102	0.897000	-3.69994×10^{-5}
0.170	0.385419	0.020166	0.370875	-4.65154×10^{-5}
0.190	0.109379	0.003641	0.107625	-2.57290×10^{-5}
0.199	0.010171	0.001048	0.010075	-3.15817×10^{-6}
0.201	-0.010007	0.001045	-0.009925	3.29943×10^{-6}
0.220	-0.171046	0.003878	-0.171000	9.69657×10^{-5}
0.260	-0.355720	0.008996	-0.357000	5.49437×10^{-4}
0.300	-0.364899	0.011647	-0.375000	1.33006×10^{-3}

where the parentheses now include both QMC and discretization errors. In the original core, all eight points lie within 0.87 combined standard deviations of (54), with $\chi^2 = 2.24/8$. In the merged sixteen-point table the maximum absolute diagonal pull is 1.31. Treating the discretization estimate as the correlated nuisance used in the fit gives $\chi^2 = 5.61/16$. The matrix-model coefficient vector $(6, -55, 150, -125)$ differs from the target-free fit by at most 14.4% in an individual coefficient; every coefficient difference is below 0.88 of its marginal fit standard error, and the full correlated coefficient quadratic form is 4.47 for four directions. These are useful diagnostics rather than formal likelihood statistics. They show good consistency while also preserving the conditioning caveat above. The complete sixteen-point table, fit covariance, and comparison grid are stored in the machine-readable artifacts cited in Appendix A.

Figure 3 shows separately the result inferred from imaginary-energy worldsheet data and the later comparison curve. Their agreement throughout the chamber, including the independently resolved zero crossing near $t = 0.2$ and the non-monotonic shape of Q_5 , tests more than an overall normalization.

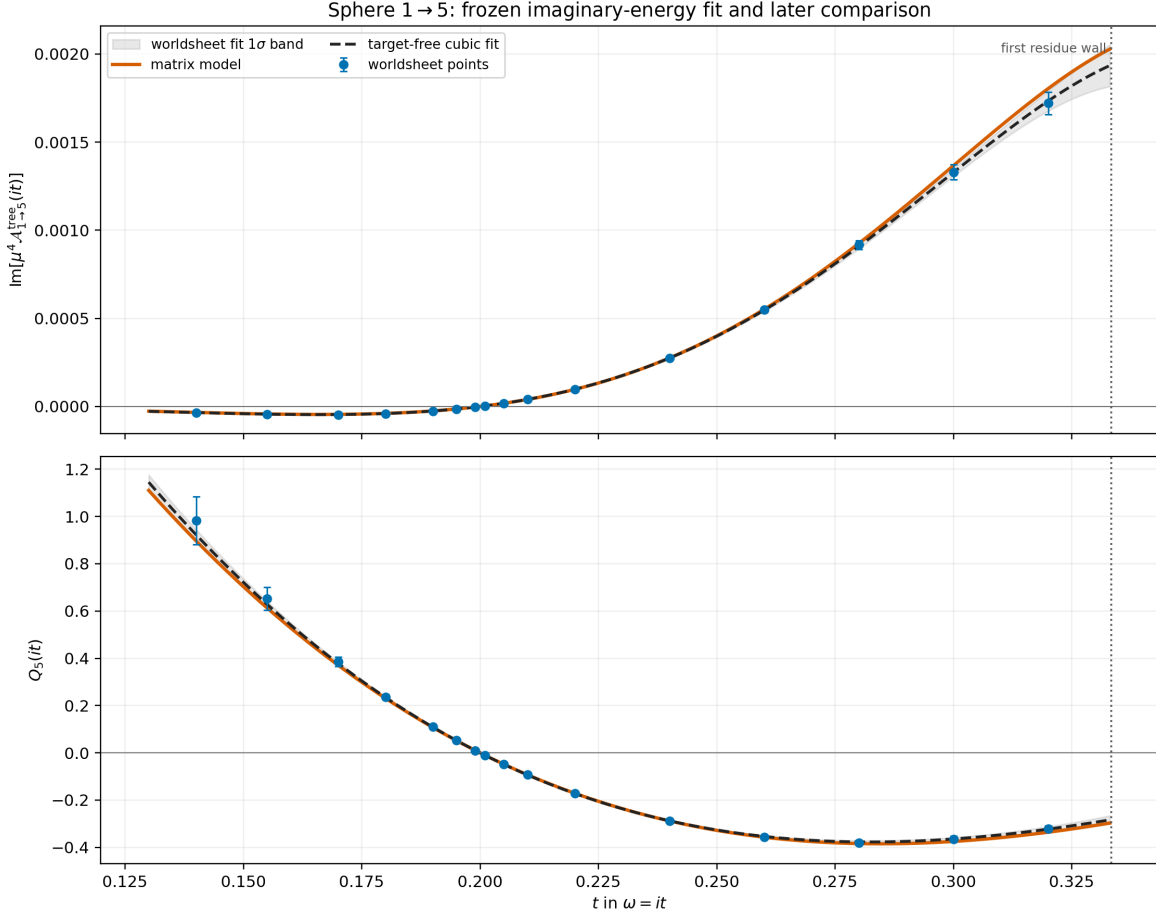


Figure 3: Sphere 1 $\rightarrow$ 5 amplitude at all sixteen local points in the residue-free chamber. Blue circles are the frozen worldsheet values, the black dashed curve and grey band are the target-free cubic fit and its one-standard-error band, and the orange curve is the matrix-model prediction loaded only afterward. The upper panel shows the imaginary part of the normalized amplitude; the lower panel shows Q_5 . Displayed point errors combine independent scrambled-Sobol QMC errors with (48) in quadrature. The dotted line marks the first Liouville residue wall at $t = 1/3$; no data are plotted beyond it.

7 Matrix-model pattern after the three comparisons

After the three process-specific worldsheet calculations and audits have been fixed, their matrix-model comparison functions can be written in the common form

$$Q_n^{\text{expected}}(\omega_{\text{in}}; \omega_1, \dots, \omega_n) = \prod_{m=1}^{n-2} (i\omega_{\text{in}} + m). \quad (57)$$

For equal outgoing energies, $\omega_{\text{in}} = n\omega$, this gives

$$\mu^{n-1} \mathcal{A}_{1 \rightarrow n}^{\text{tree}}(\omega) = i n \omega^{n+1} \prod_{m=1}^{n-2} (m + i n \omega). \quad (58)$$

Equations (57) and (58) summarize the comparison target supported by the $1 \rightarrow 3$, $1 \rightarrow 4$, and $1 \rightarrow 5$ data. They are not inputs to any of the worldsheet integrations and do not constitute a worldsheet proof at arbitrary multiplicity.

8 A scalable route to general $1 \rightarrow n$

A sphere $1 \rightarrow n$ amplitude is an $(n+1)$ -point correlator. A trivalent channel contains

$$(n+1) - 3 = n - 2 \quad (59)$$

internal Liouville momenta. A scalable computation should separate the discrete contour bookkeeping from the numerical contraction of the continuous integrals.

8.1 Momentum-contour forest

For every trivalent vertex:

1. construct the affine DOZZ pole hyperplanes in the adjacent internal momenta and external energies;
2. determine which hyperplanes cross the declared contours as the external energies move;
3. add the residues with their oriented signs;
4. inspect each residue integrand recursively for further crossings;
5. organize simultaneous crossings as lower-dimensional residue strata.

Within a fixed kinematic chamber, the answer becomes a sum of integrals of dimensions

$$n - 2, n - 3, \dots, 0. \quad (60)$$

This “contour forest” is the higher-point version of the single residue in (29).

8.2 Quadrature and tensor contraction

Each continuous or residue stratum should use:

- a Gauss–Jacobi threshold panel after the exact small- P power has been factored out;
- ordinary bulk panels divided at nearby pole projections and pole widths;
- a separate Gaussian-tail or rationally mapped tail panel;
- a threshold block evaluator that avoids nearly degenerate fixed-weight c -recursion nodes.

At fixed plumbing coordinates, the trivalent tree is naturally a tensor network. Momentum nodes are edge indices, DOZZ and descendant couplings are vertex tensors, and plumbing powers are diagonal edge tensors. Contracting the tree locally avoids an explicit M^{n-2} tensor grid. Without further compression, the generic cost is of order

$$O(nM^3). \tag{61}$$

9 Error budget and limitations

For every future $1 \rightarrow n$ production point, the following uncertainties should be retained separately:

1. adjacent momentum rules and threshold refinements;
2. adjacent Virasoro-block orders;
3. moduli QMC depth and independent scrambles;
4. agreement between overlapping OPE channels;
5. distance to the next momentum-contour wall;
6. residue-stratum quadrature;
7. any rational-reconstruction or degree assumption.

The present $1 \rightarrow 3$, $1 \rightarrow 4$, and $1 \rightarrow 5$ evidence identifies the corresponding analytic answers very strongly in controlled imaginary-energy chambers. It is not, by itself, a theorem that Q_4 has degree two or that Q_5 has degree three. Nor does it yet constitute a worldsheet derivation of (57) for arbitrary n .

A Machine-readable provenance and implementation map

This appendix is a reviewer-oriented map of the code that directly supports the sphere $1 \rightarrow n$ claims. File names are relative to the `plumbing/` directory. The self-contained review snapshot is generated by `build_sphere_one_to_n_review_bundle.py`; its manifest follows all local Python imports and therefore includes shared transitive dependencies not listed individually below.

A.1 Trust boundary used in every blind calculation

The production flow is deliberately split into files with different authority:

1. a worksheet kernel evaluates Liouville structure constants, timelike factors, conformal blocks, and moduli weights;
2. a worksheet scan produces raw integrals and stripped amplitudes but contains no target polynomial;
3. an audit varies numerical settings and is also target-blind;
4. the worksheet JSON is hashed and frozen;
5. only a separate comparison file imports or evaluates the matrix-model function, after verifying the frozen hashes.

This boundary is enforced explicitly for all three cases. In the current $1 \rightarrow 4$ path, the twelve-point extension is produced without a target polynomial, the merger verifies the historical points and extension hashes, and the target-free fitter freezes the thirty-point table and twenty-nine-point primary fit. Only then does the downstream comparison verify those hashes and evaluate the matrix coefficients. The earlier eighteen-point fit is retained only as input-lineage provenance and is not a competing result in this note. The $1 \rightarrow 5$ path now applies the same separation to its sixteen residue-free imaginary-ray points and cubic fit.

A.2 Shared normalization, Liouville input, and blocks

Normalization

`audit_genus0_one_to_two_amplitude.py` derives the sphere normalization from the three-point anchor; `audit_genus0_one_to_two_amplitude_checks.py` is its regression gate. `audit_c1_sphere_topology_normalization.py` and its check file provide an independent topology-normalization ledger.

Liouville data

`liouville_torus.py` implements Υ_1 , the logarithmic delta-normalized DOZZ/Yin structure constant, and shared special-function conventions. `liouville_torus_checks.py` tests reflection, shift, and normalization identities. No matrix-model information enters this layer.

Virasoro blocks

`ccy_sphere_four_point.py`, `ccy_sphere_five_point.py`, `ccy_sphere_six_point.py`, and `ccy_sphere_six_point_star.py` implement the four-, five-, and six-point CCY recursions in the channel coordinates used by the moduli atlases. Each has a corresponding `*_checks.py` file. The checks compare low levels with direct descendant sums, verify elliptic-nome re-expansion, exchange symmetries, and comb-star factorization limits.

A.3 Sphere $1 \rightarrow 3$ code path

Convergent kernel `sphere_four_point_imaginary_energy.py` fixes $\omega = it$, constructs the one-momentum Liouville kernel, and implements the exact three-chart proposal

(15).

Blind production `sphere_four_point_worldsheet_scan.py`

evaluates both the original sixteen-point table and the fourteen-point interlaced extension, records all replicate estimates and settings, and writes their frozen hashes.

Blind merge and fit

`merge_sphere_four_point_worldsheet_scans.py` verifies the original and extension hashes, common numerical design, disjoint seed streams, and interlaced kinematics before freezing the thirty-point table. `sphere_four_point_imaginary_ray_fit.py` then freezes the primary, cohort, and deep-audit-replacement affine fits without importing a target coefficient.

Blind convergence `sphere_four_point_worldsheet_audit.py` verifies the production hash before varying block order, momentum order, cutoff, and RQMC depth. It produced a four-point audit and a separate deep audit at $t = 0.32$.

Later comparison `sphere_four_point_30point_matrix_comparison.py`

contains the new thirty-point Q_3^{expected} comparison. It refuses mismatched scan or fit hashes and produces the cohort diagnostics and `amplitude_comparison_30point.png`.

The earlier `sphere_four_point_matrix_comparison.py` remains the frozen sixteen-point comparison record.

Physical control branch

`sphere_four_point_physical.py` and `sphere_four_point_subtraction.py` implement the independent physical *ie* BRY finite part. Their check files verify collar independence, local projector cancellation, energy conservation, and the published four-point benchmark. This branch shares block and DOZZ primitives but is not used to generate the imaginary-ray production table.

The inherited audit artifacts are `results/sphere_four_point_1to3/worldsheet_audit.json` and `results/sphere_four_point_1to3/worldsheet_audit_t032.json`. The new thirty-point artifacts reside in

`results/sphere_four_point_1to3/blind30_20260824/`

Its principal files are

- `worldsheet_extension_14point.json`;
- `worldsheet_scan_30point.json`;
- `worldsheet_affine_fit_30point_frozen.json`;
- `matrix_comparison_30point.json`.

A.4 Sphere $1 \rightarrow 4$ code path

Geometry and Liouville

`sphere_five_point_liouville.py` constructs labelled five-point channels, their Jacobians, and Liouville factors; `sphere_five_point_liouville_checks.py` tests channel overlap and covariance.

Main kernel and atlas

`sphere_five_point_equal_energy.py` precomputes the double-momentum kernel, evaluates the 120 oriented plumbing charts, implements the exact atlas-mixture density, and contains both the subtraction-free and physical forest estimators.

Contour continuation

`sphere_five_point_extended_scan.py` evaluates the imaginary ray with the continued real contour and the $-2i$ crossed-pole residue above $t = 2/5$. `sphere_five_point_continuation_checks.py` tests the residue and its chamber of validity.

Physical finite part

`sphere_five_point_subtraction.py`, `sphere_five_point_physical_scan.py`, and their check files implement and test the physical BRY forest independently of the convergent-ray scan. This branch remains a smoke test and supplies no curve used in this note.

Thirty-point production and fit

`sphere_five_point_30point_worldsheet_extension.py`
computes and freezes the twelve new points with disjoint Sobol seed blocks.
`sphere_five_point_30point_worldsheet_fit.py`
verifies the historical table and extension hashes, merges all thirty points, and performs the target-free quadratic fits. Neither file contains a matrix-model coefficient.

Worksheet audit

`sphere_five_point_30point_audit_summary.py` freezes the independent block-order checks at $t = 0.31, 0.35, 0.44$ and constructs the conservative local uncertainties. The full regression and lineage gate is `sphere_five_point_30point_checks.py`.

Later comparison

`sphere_five_point_30point_matrix_comparison.py`
accepts only the frozen thirty-point scan, fit, and audit, verifies their hashes, and then evaluates the matrix-model coefficients and produces Figure 2.
`sphere_five_point_numerical_recipe.md`
records the two distinct imaginary-ray and direct-physical prescriptions.
`sphere_n_point_momentum_integration.md`
records the general momentum strategy.

The only current $1 \rightarrow 4$ result artifacts used in this note reside in

`results/sphere_five_point_1to4/blind30_20260824/`

Their principal files are

- `worldsheet_extension_12point.json`;
- `worldsheet_scan_30point.json`;
- `worldsheet_points_30point_frozen.json`;
- `worldsheet_quadratic_fit_30point_frozen.json`;
- `worldsheet_newpoint_block_audit_frozen.json`;
- `matrix_comparison_30point.json`;
- `amplitude_comparison_30point.png`.

A.5 Sphere $1 \rightarrow 5$ code path

Six-point atlases	<code>sphere_six_point_atlas.py</code> enumerates 720 labelled comb and 720 labelled star charts, maps their plumbing variables to puncture positions, computes Jacobians, and evaluates the full mixture density. <code>sphere_six_point_atlas_checks.py</code> verifies coverage, label counting, Jacobians, and overlap transformations.
Worksheet kernel	<code>sphere_six_point_equal_energy.py</code> constructs the cached three-momentum DOZZ tensor, selects the appropriate comb or star block, evaluates the time-like factor, and contracts the block at every moduli point. <code>sphere_six_point_equal_energy_pilot.py</code> fixes the first wall and benchmarks block cost; it is not a production substitute.
Production/audit	<code>sphere_six_point_worldsheet_scan.py</code> is the target-blind production driver. <code>sphere_six_point_worldsheet_audit.py</code> performs momentum, block, cut-off, radial-sampler, and channel diagnostics without importing the prediction.
Extension/merge	The same scan driver produced the later eight local points. <code>merge_sphere_six_point_worldsheet_scans.py</code> verifies distinct points, chamber membership, zero fallback counts, and settings compatibility before writing and freezing the sixteen-point merged table.
Imaginary-ray fit	<code>sphere_six_point_imaginary_ray_fit.py</code> verifies the merged scan and its original freeze, writes a target-free points-only table, and performs the cubic GLS fit in (49). Its regression and trust-boundary gate is <code>sphere_six_point_imaginary_ray_fit_checks.py</code> .
Later comparison	<code>sphere_six_point_imaginary_ray_fit_comparison.py</code> verifies the frozen fit and its source-point hash before evaluating the expected Q_5 polynomial and producing Figure 3. <code>sphere_six_point_matrix_comparison.py</code> retains the earlier direct pointwise comparison as an independent consistency artifact.
Cannon campaign	<code>sphere_six_point_cannon_blind.py</code> , <code>sphere_six_point_cannon_compare_after_freeze.py</code> , their checks, the <code>config/sphere_six_point_1to5_cannon_blind50_v1.json</code> configuration, and the <code>cluster/sphere_six_point_1to5_blind_*.slurm</code> scripts are infrastructure for the requested 50-point high-accuracy campaign. That campaign was cancelled. No queued or partial

Cannon value is used in any table, fit, figure, or conclusion in this note. A replacement blind 30-point design, `config/sphere_six_point_1to5_cannon_blind30_3h_v1.json`, has now been submitted. It uses 14 production replicates of 2^{15} Sobol points and a 5×10^{-4} absolute Q_5 gate. A strict three-hour preflight (job 41508113) submitted no production work: the production benchmark projected 1.72 hours before its safety factor, and the systematics benchmark reached its 20-minute cap during high-order kernel construction. Without changing any numerical setting, the production wall limits were relaxed to 10 hours for the array, 30 minutes for assembly, and 20 minutes for validation. The active job chain is array 41514317, assembly 41514333, and freeze 41514344. Until the freeze gate succeeds, this campaign contributes no numerical value to the present conclusions.

The principal $1 \rightarrow 5$ artifacts are

- `results/sphere_six_point_1to5/worldsheet_convergent_scan.json`;
- `results/sphere_six_point_1to5/worldsheet_convergent_scan_extension8_local.json`;
- `results/sphere_six_point_1to5/worldsheet_convergent_scan_16point_local.json`;
- `results/sphere_six_point_1to5/worldsheet_freeze_manifest_16point_local.json`;
- `results/sphere_six_point_1to5/worldsheet_numerical_audit.json`;
- `results/sphere_six_point_1to5/worldsheet_imaginary_ray_points_frozen.json`;
- `results/sphere_six_point_1to5/worldsheet_imaginary_ray_cubic_fit_frozen.json`;
- `results/sphere_six_point_1to5/matrix_model_fit_comparison_16point_local.json`.

A.6 Recommended faithfulness-review order

The shortest review path that still checks every conceptual boundary is:

1. verify the labelled normalization with `audit_genus0_one_to_two_amplitude.py`;
2. inspect `liouville_torus.py` and its special-function checks;
3. run the four-, five-, and six-point CCY block checks;
4. inspect the exact proposal-mixture densities in the $1 \rightarrow 3$ kernel, five-point equal-energy kernel, and six-point atlas;
5. trace one integrand from external energies through DOZZ tensors, block evaluation, timelike factor, Jacobian, and proposal weight;
6. verify the frozen JSON hashes independently;
7. confirm by source inspection that worldsheet scan and audit files do not import comparison modules or contain target polynomials;
8. only then run the three matrix-comparison files.